

Giacomo Galloni

Postdoctoral Researcher — Cosmology · University of Ferrara

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RESEARCH INTERESTS

My research focuses on cosmic inflation through the Cosmic Microwave Background (CMB) and Gravitational Waves (GWs), with a methodological emphasis on likelihood analysis. I lead the large-scale CMB likelihood effort within the LiteBIRD collaboration and co-lead the Likelihood & Theory group of the Simons Observatory (SO), where I develop end-to-end pipelines that combine harmonic and pixel-based estimators with realistic component separation. In parallel, I develop CosmoForge, a unified likelihood framework for current and future CMB experiments, which I keep improving and extending with new features. On the science side, I have produced Bayesian and frequentist constraints on the amplitude and tilt of the primordial tensor spectrum, and I work on stochastic GW backgrounds and their cross-correlation with the CMB, including recent data-driven bounds on the quantum decoherence of inflationary GWs.

POSITIONS & EDUCATION

Feb 2024 — present	Postdoctoral Researcher , University of Ferrara.
Nov 2020 — Jan 2024	Ph.D. with honors in Astronomy, Astrophysics and Space Science , University of Roma Tor Vergata. Supervisors: Prof. Nicola Vittorio, Prof. Sabino Matarrese.
Oct 2018 — Oct 2020	M.Sc. with honors, Physics of the Universe , University of Padova. Thesis: Prof. Nicola Bartolo, Dr. Angelo Ricciardone.
Oct 2014 — Oct 2018	B.Sc. in Physics , University of Padova. Thesis: Prof. Giacomo Ciani.

FELLOWSHIPS & VISITINGS

Sep — Dec 2023	Funded visiting at IJCLab, Orsay (France).	Nov 2020 — Jan 2024	Full Ph.D. fellowship, University of Roma La Sapienza.
Jun — Jul 2023	Funded visiting at Kavli IPMU, Tokyo (Japan) via MSCA CMB-INFLATE.	Jun — Sep 2017	Funded internship at the University of Florida (bachelor thesis).

SELECTED PUBLICATIONS

Full list on [Google Scholar](#) and [INSPIRE-HEP](#). First-author and small-group papers below.

- Galloni, G. and L. Pagano (2026). “*CosmoForge I: A unified framework for QML power spectrum estimation and pixel-based likelihood analysis*”. arXiv: 2605.21149 [astro-ph.CO]
- Genesini, V., G. Galloni, et al. (2026). “*Cross-spectra likelihood for robust τ constraints from all satellite polarisation data*”. arXiv: 2603.22454 [astro-ph.CO]
- Galloni, G., P. Campeti, et al. (2025). “*Accurate and efficient likelihood modeling for large-scale CMB data*”. JCAP 12, p. 052 — doi:10.1088/1475-7516/2025/12/052 — arXiv:2505.24829
- Giardiello, S., A. J. Duivenvoorden, et al. (2025). “*Modeling beam chromaticity for high-resolution CMB analyses*”. Phys. Rev. D 111, p. 043502 — arXiv:2411.10124
- Kruijff, J. de, G. Galloni, et al. (2025). “*The first data-driven bounds on the quantum decoherence of inflationary gravitational waves*”. arXiv: 2511.14727 [astro-ph.CO]
- Galloni, G., S. Henrot-Versillé, et al. (2024). “*Robust constraints on tensor perturbations from cosmological data: a Bayesian and frequentist comparative analysis*”. Phys. Rev. D 110.6, p. 063511 — arXiv:2405.04455
- Galloni, G., M. Ballardini, et al. (2023). “*Unraveling the CMB lack-of-correlation anomaly with the cosmological GW background*”. JCAP 10, p. 013 — arXiv:2305.18184
- Galloni, G., N. Bartolo, et al. (2023). “*Updated constraints on amplitude and tilt of the tensor primordial spectrum*”. JCAP 04, p. 062 — arXiv:2208.00188
- Galloni, G., N. Bartolo, et al. (2022). “*Test of the statistical isotropy of the universe using gravitational waves*”. JCAP 09, p. 046 — arXiv:2202.12858

INDEPENDENT PROJECTS

Ongoing	CosmoForge — Power Spectrum and Likelihood Framework. Python framework for power-spectrum estimation and likelihood analysis on large-scale CMB data, designed as a unified tool for LiteBIRD, SO, and future experiments.
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In prep.

BICEP Diagnosis with the marginalized HL Likelihood. Re-analysis of BICEP data exploiting new capabilities of the marginalized Hamimeche & Lewis (mHL) likelihood, stress-testing existing constraints on tensor modes.

COLLABORATION PROJECTS

<i>Jun 2024</i> —	LiteBIRD — Foreground and Sensitivity Studies. Interfacing my Python package with foreground separation and sensitivity analyses.
<i>May 2024</i> —	SO — Likelihood & Theory. Co-lead of the group; maintenance and development of SOLikeT. Contributed single-mode likelihoods and led the restructuring of LAT MFLike.
<i>Sep 2024</i> —	SO — Power Spectrum. Contributing to the Power Spectrum group, with planned developments on QML-based estimators for large-scale modes.
<i>Jan 2026</i> —	LiteBIRD — Moment-Expansion Component Separation. Improving and characterizing the moment-expansion approach in the LiteBIRD context.
<i>Jun 2023</i> —	LiteBIRD — MaPLEs team. Leading efforts to study and characterize the likelihood of LiteBIRD; released within the collaboration a Python package to perform this analysis.
<i>Apr 2023</i> —	LiteBIRD — Simulation Team. Validation tests, cosmological analysis of CMB temperature with LiteBIRD, and development of parts of <code>litebird_sim</code> .
<i>May 2022</i> —	LiteBIRD — Tests on Cosmic Inflation. Monte-Carlo study of which inflationary models can be detected by LiteBIRD.

SERVICE & LEADERSHIP

<i>Sep 2024</i> —	Co-lead, Likelihood & Theory group, Simons Observatory.	<i>Feb 2026</i> —	Member, Interim Governance Board, LiteBIRD.
<i>Dec 2025</i> —	Early Career Coordination (ECC) representative, LiteBIRD.	<i>Jan 2026</i>	SOC member, LiteBIRD Face-to-Face Meeting.
		<i>Jul 2024</i>	Session organizer & chair on the present/future of CMB, Marcel Grossman XVII (Pescara).

SELECTED TALKS

<i>Jul 2026</i>	Highlight Talk, Simons Observatory Collaboration Meeting, Toronto (Canada).	<i>May 2025</i>	Invited Outreach Talk, Pint of Science, Ferrara (Italy).
<i>Jul 2026</i>	Invited Review Talk, CMB Challenges, Chino (Japan).	<i>Dec 2024</i>	Invited Seminar, University of Rome Tor Vergata (Italy).
<i>May 2026</i>	Invited Seminar at SISSA (Italy).	<i>Dec 2024</i>	Invited Seminar, University of Padova (Italy).
<i>Apr 2026</i>	Invited Speaker, GRaviCon 2026, Pisa (Italy).	<i>Jul 2024</i>	Invited Speaker, Marcel Grossman XVII, Pescara (Italy) — ICRANet.
<i>Nov 2025</i>	Invited Speaker, GW-CMB 2025, National Central University (Taiwan).	<i>Jul 2024</i>	Invited Speaker, KEK CMB Group Meeting, Tsukuba (Japan).
<i>Sep 2025</i>	Invited Speaker, Tensions in Cosmology, Corfu (Greece).	<i>Feb 2024</i>	Invited Speaker, ASI-LiteBIRD, Frascati (Italy).

TEACHING & MENTORSHIP

<i>a.y. 2026/27</i>	Bayesian Statistics , M.Sc. in Physics, University of Ferrara. Co-taught with M. Lattanzi.
<i>Feb 2026</i> —	External mentor of a Ph.D. student at National Central University (Taiwan), working on $r-n_t$ constraints from a joint analysis of LIGO-Virgo-KAGRA likelihoods and Simons Observatory forecasts.
<i>Nov 2025 — Jul 2026</i>	Co-supervisor of a Master student at the University of Padova, on the recovery of r and n_t from SO forecasts.
<i>2024</i>	Mentor of a Ph.D. student at the University of Padova, on the first data-driven bounds on the quantum decoherence of inflationary GWs.
<i>Feb 2024</i> —	Supervisor, then co-supervisor with L. Pagano, at the University of Ferrara: Master thesis on τ from Planck and WMAP, then a Ph.D. on a cross-spectra likelihood for τ (Genesini et al. 2026), moment-expansion component separation and reionization history.

SOFTWARE SKILLS

Statistics. Expert in Bayesian and frequentist methods. Author of CosmoForge, a unified Python framework for QML and pixel-based CMB likelihood analyses; contributed to Cobaya; developed a profile-likelihood sampler.

Programming. Highly proficient in Python and C, with strong experience in package design, maintenance, and CI workflows. Author of CosmoForge, maintainer of SOLikeT, contributor to `litebird_sim` and several open-source cosmology repositories.

Boltzmann codes. Proficient user of CAMB and CLASS.